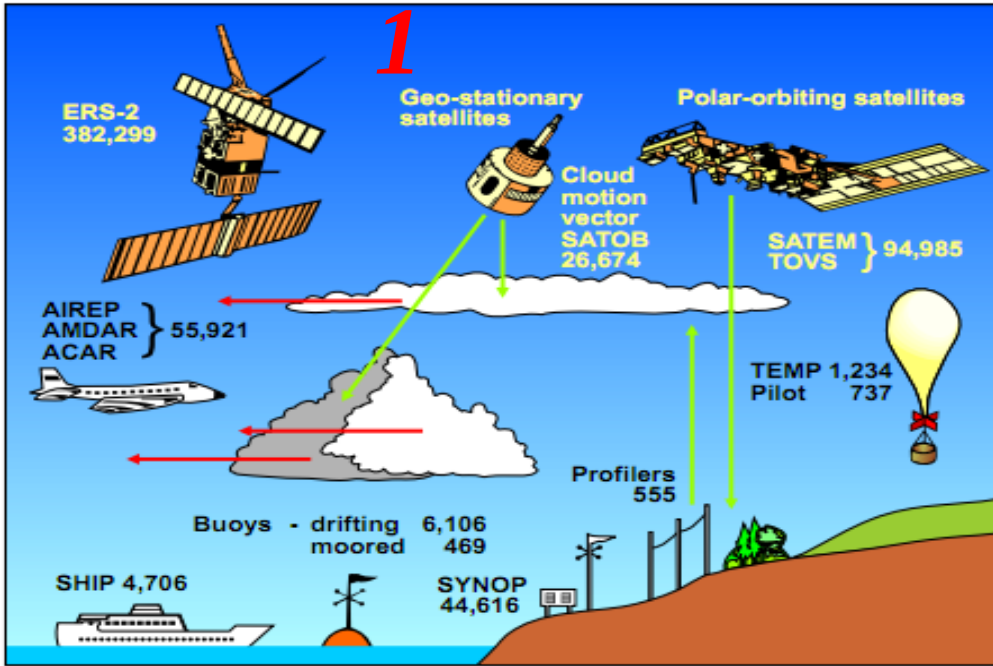


Introduction to Numerical Weather Prediction



3
Weather Prediction Equation

4
Numerical Methods



Introduction to Numerical Weather Prediction



Chapter 1: Introduction



Objective

- *To Identify Different Weather Forecasting Technique*
- *To Define Numerical Weather Prediction*
- *To Know The Historical Development Of NWP*

Introduction

- The story goes that in 1904 Wilhelm Bjerknes's was the first to point out that the future state of the atmosphere could be predicted by integrating the partial differential equations that govern the behavior of the atmosphere, using as initial fields the observed state of the atmosphere at a particular time.

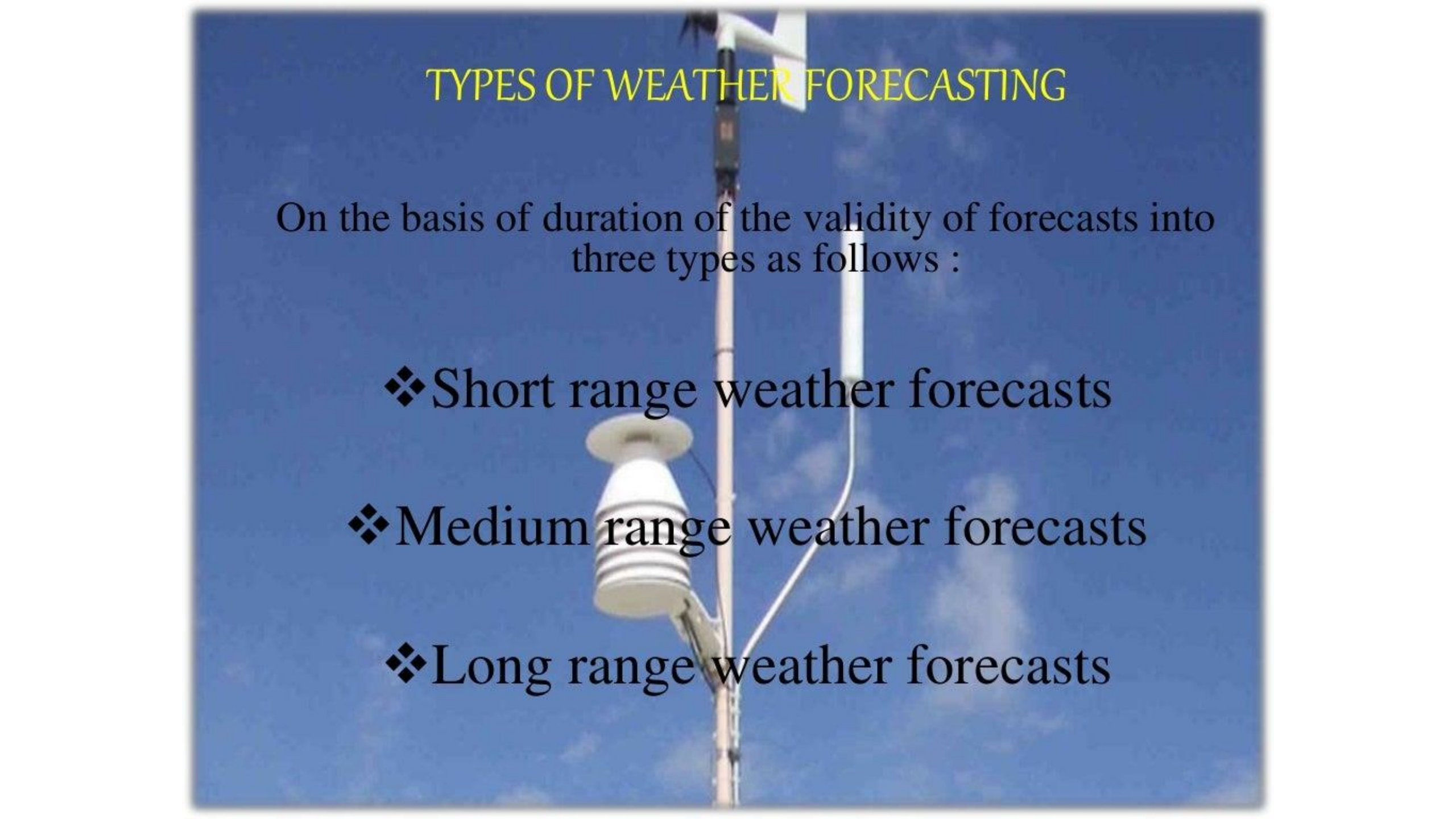
Cont..

- However, the equations are too complicated for analytic solutions to be found and one must resort to numerical methods.
- We refer now to such integrations as numerical weather prediction, commonly abbreviated NWP.

Weather forecasting

- is the prediction of what the atmosphere will be like in a particular place by using technology and scientific knowledge to make weather observations.
- In other words, it's a way of predicting things like cloud cover, rain, snow, wind speed, and temperature before they happen.

TYPES OF WEATHER FORECASTING



On the basis of duration of the validity of forecasts into three types as follows :

- ❖ Short range weather forecasts
- ❖ Medium range weather forecasts
- ❖ Long range weather forecasts

Short-range weather forecasts

- Most useful predictions of weather wherein forecasts valid from few hours to 48 hours and sometime 72 hours
- Based on maps weather charts , satellite images
- Determines change of atmospheric weather of a specific location
- Persistent method & continuity methods are applied .
- Helps in transportation and fishermen

Medium range weather forecasts

- ❖ It covers the time span from 3 days to 3 weeks
- ❖ This prediction is helpful for farmers
- ❖ This prediction is done by calculating the average of past and present weather condition

Long range weather forecasts

- ❖ It covers a time span from a fortnight to a season of the year
- ❖ These are basically statements ,exact accuracy is minimum
- ❖ Not so detailed information

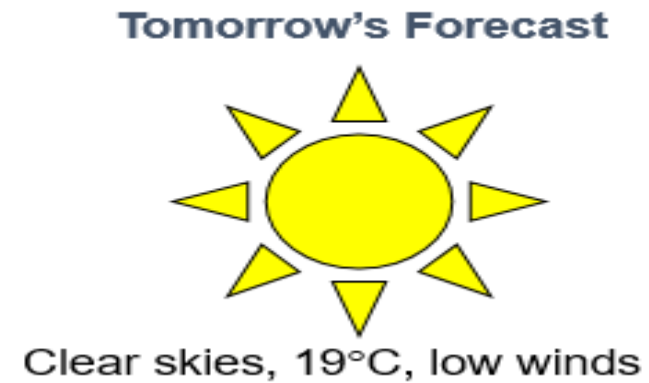
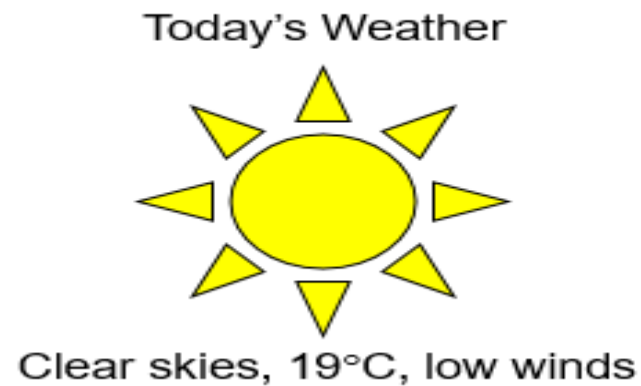
Types of Forecasts

- Persistence
- Trend
- Analog
- Climatology
- Numerical Weather Prediction

Types of Forecasts

Persistence - forecast the future atmospheric state to be the same as current state

- Raining today, so forecast rain tomorrow
- Useful for few hours to couple days



Types of Forecasts

Trend - add past change to current condition
to obtain forecast for the future state

-Useful for few hours to couple days

<u>10 am</u>	<u>11 am</u>	<u>12 pm</u>
19° C	23° C	27° C
Past	Now	Future

Types of Forecasts

Analog - find past state that is most similar to current state, then forecast same evolution

- The analog method is a difficult method to use when predicting the weather because it requires finding a day in the past with weather similar to the current forecast, which is difficult to do.
- Useful for forecasts up to one or two days

Types of Forecasts

Climatology - forecast future state to be same as climatology or average of past weather for date

- Forecast July 4th MAX for Arba Minch to be 25°C
- Most accurate for long forecast projections,
forecasts longer than 30 days

Types of Forecasts

NWP:- use mathematical models of physics principles to forecast future state from current conditions.

- The technique used to obtain an objective forecast of the future weather (up to possibly two weeks) by solving a set of governing equations that describe the evolution of variables that define the present state of the atmosphere.

NWP

- How can “numerical weather prediction” be defined?
- **Numerical:** using data or numbers; in reference to the methods used (i.e., the solving of mathematical equations).

NWP

- **Prediction:** an act of declaration of a future event; more generally, a forecast of a past, present, or future event.
- **Weather:** the state of the atmosphere with respect to wind, temperature, moisture, pressure, etc.; also referring to the time scale (e.g., in contrast to climate).

What is NWP?

What is NWP ?

When a **numerical model** is used in meteorology to forecast the **weather** the process is called **Numerical Weather Prediction (NWP)**.

A **numerical model** is a set of computer programs designed to simulate a real system; in our case weather.

Equations based on laws of physics are integrated forward in time to simulate changes in the atmosphere.

What is NWP?

- **Numerical weather prediction (NWP)** is a method of weather forecasting that employs a **set of equations** that describe the flow of fluids.
- These equations are translated into **computer code** and use **governing equations, numerical methods, parameterizations** of other physical processes and combined with **initial and boundary** conditions before being run over a domain (geographic area).

What is NWP?

- NWP uses the latest weather observation alongside a mathematical computer model of the atmosphere (such as the Unified model run at the Met Office) to produce a weather forecast.
- NWP is now *at the heart of all weather* forecasts and *warnings* as well as much of the *research* and development undertaken to understand our weather and climate.
- A numerical weather model is a computer program that simulates the atmospheric motion in space and time. A variety of weather phenomena can be analyzed and predicted by these types of NWP models.

What is NWP?

- At the heart of any major weather forecasting center is a numerical weather prediction (NWP) model – a computerised model of the atmosphere which, given an initial state for the atmosphere, derived from observations, can generate forecasts of how the weather will evolve into the future.
- Numerical weather prediction involves the use of mathematical models of the atmosphere to predict the weather. Manipulating the huge datasets and performing the complex calculation

- Numerical weather prediction models are equations describing the dynamical behavior of the atmosphere, typically coded into a computer program.

Historical Development Of Numerical Weather Prediction (NWP)

Historical Development of NWP

- ❖ The roots of numerical weather prediction can be traced back to the work of Vilhelm Bjerknes, a Norwegian physicist who has been called the father of modern meteorology.
- ❖ V. Bjerknes in 1904 was the first to point out that the future state of the atmosphere could be predicted by integrating the partial differential equations that govern the behavior of the atmosphere, using the observed state of the atmosphere at a particular time as initial fields.
- ❖ In 1904, he published a paper suggesting that it would be possible to forecast the weather by solving a system of nonlinear partial differential equations.



Historical Development of NWP

- In 1904, Vilhelm Bjerknes derived a two-step procedure for model-based weather forecasting. First, a diagnostic step is used to process data to generate initial conditions, which are then advanced in time by a prognostic step that solves the initial value problem.
- He also identified seven variables that defined the state of the atmosphere at a given point: *pressure, temperature, density, humidity, and the three components of the flow velocity vector.*
- He proposed that, given the initial atmospheric condition, it is possible to compute the future state of the atmosphere using mathematical formula.

Historical Development of NWP

- Bjerknes pointed out that equations based on mass continuity, conservation of momentum, the first and second laws of thermodynamics, and the ideal gas law could be used to estimate the state of the atmosphere in the future through Numerical methods.
- With the exception of the second law of thermodynamics, these equations form the basis of the primitive equations used in present-day weather models.

Historical Development of NWP

- ❑ After Bjerknes, A British mathematician named Richardson (1922) made the first attempt at NWP by hand. He used full primitive equations and a finite difference scheme.
- ❑ for the state of the atmosphere over two points in central Europe, taking at least six weeks to do so.
- ❑ He divided the region of interest into cells, like the squares on a chessboard. He read the atmospheric conditions from a weather map using manual interpolation.
- ❑ Richardson predicted a “forecast factory,” where he calculated that 64,000 human “computers,” each responsible for a small part of the globe, would be needed to keep “pace with the weather” in order to predict weather conditions.



Historical Development of NWP

- They would be housed in a circular hall like a theater, with galleries going around the room and a map painted on the walls and ceiling.
- A conductor located in the center of the hall would coordinate the calculations using colored lights...
- At that time the calculations had to be carried out by hand and were very tedious and time-consuming.
- The calculation took him six weeks, and the prediction turned out to be completely unrealistic, but his efforts were a glimpse into the future of weather forecasting.

Historical Development of NWP

- Even though his methodology was perfect, he obtained the forecast change in surface pressure at 145 mb in 6 h! The failure of Richardson's forecast set the NWP concept back into the theoretical world for many years.
- The large error was caused by an imbalance in the pressure and wind velocity fields used as the initial conditions in his analysis.
- While Richardson's vision never became a reality, the use of mathematics to predict the weather did develop over the years.



This illustration depicts Richardson's
"forecast factory."

An artist's impression of Richardson's Forecast Factory



Today's super computers ~ 10 billion of Richardson's Forecast Factories

Historical Development of NWP

- The first successful numerical prediction was performed using the ENIAC digital computer in 1950 by a team led by American meteorologist Jule Charney.
- The team include Philip Thompson, Larry Gates, and Norwegian meteorologist Rangnar Fjortoft, applied mathematician Johan van Neumann, and computer programmer Klara Dan Von Neumann, M. H. Frankel, Jerome Namias, John C. Freeman Jr., Francis Reichelderfer, George Platzman, and Joseph Smagorinsky.



Figure 1: The ENIAC computer in 1948. The operators are changing the plug-in wiring. (PLATZMAN, 1979).

Historical Development of NWP

- They used a simplified form of atmospheric dynamics based on solving the barotropic vorticity equation over a single layer of the atmosphere, by computing the geopotential height of the atmosphere's 500 millibars (15 inHg) pressure surface.
- This simplification greatly reduced demands on computer time and memory, so the computations could be performed on the relatively primitive computers of the day.

Historical Development of NWP

- When news of the first weather forecast by *Electronic Numerical Integrator and Calculator* (ENIAC) was received by Richardson in 1950, he remarked that the results were an “enormous scientific advance”.
- The first calculations for a 24-hour forecast took ENIAC nearly 24 hours to produce, but Charney's group noted that most of that time was spent in "manual operations", and expressed hope that forecasts of the weather before it occurs would soon be realized.

Historical Development of NWP

- ❑ The prediction was based on the integration of the absolute vorticity conservation equation.
- ❑ In the 1960s, increasing computer power made it possible to go beyond regional weather simulations to model the global general circulation.
- ❑ This, in turn, allowed scientists to simulate climate, or the average state of the atmosphere over long periods (decades to centuries).

Historical Development of NWP

- ❑ By the 1970s, general circulation models (GCMs) had become the central tool of climate science. GCM simulations provided a crucial means of analyzing the effects of climate change.
- ❑ Since the mid-1980s coupled OAGCMs (Ocean Atmospheric General Circulation Models), had established a new standard for climate modeling.
- ❑ In the last four decades, progress in the accuracy and sophistication of NWP models has been swift, partly through the development of *improved numerical algorithms for solving the equations and also because of the astonishing technological developments in computer field.*

❑ Ensemble forecasting: Instead of a single deterministic forecast, ensemble forecasting has become the mainstream method of operational NWP today (see Kalnay 2003; Bauer et al. 2015).

- ❑ Today, NWP has become a multidisciplinary science in both research and operational environments. Many countries have their own NWP systems for daily operational forecasting, including both global and regional model systems for NWP.
- ❑ In addition, NWP computer models have become useful tools for research and education.
- ❑ The development of community weather and climate models, led by the US institutes, e.g., National Center for Atmospheric Research (NCAR), Penn State University, etc.
- ❑ makes these weather and climate research and forecasting models available to many research institutes and universities around the world.

- ❑ Along with the rapid development in computer power and computer science during the last 60 years, NWP skill has been steadily improved.
- ❑ Figure 1 shows the forecast skill improvements at ECMWF between 1981 and 2014 (note that other centers follow a similar trend of improvement).
- ❑ It is clear that NWP products are quite reliable within a 5-day range and useful in a 7-day range.
- ❑ The improvement attained since the late 1990s from the advanced use of satellite observations is remarkable, especially in the Southern Hemisphere.

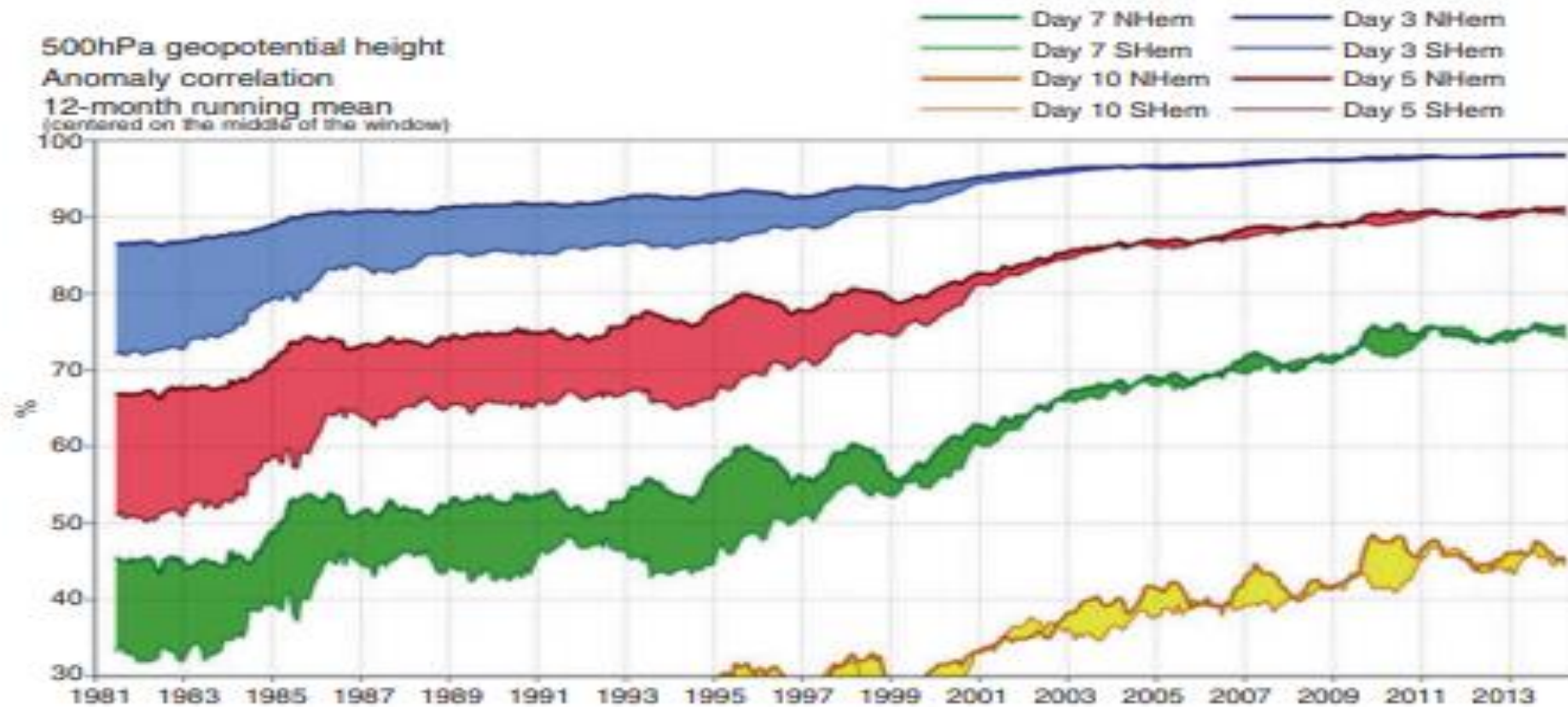


Fig. 1 Time series of the annual running mean of anomaly correlations of 500 hPa geopotential height forecasts evaluated against the operational analyses for the period of January 1981 till present. Values plotted at a particular month are averages over that month, the previous 5 months, and the following 6 months. Forecast lead times of 3, 5, 7, and 10 days are shown, for scores averaged over the northern (bold lines) and southern (thin lines) extratropics. The shading shows differences in scores between the two hemispheres at the forecast ranges indicated. (Adapted and extended from Simmons and Hollingsworth 2002. Verification follows updated WMO/CBS guidelines as specified in the Manual on the GDPFS, Volume 1, Part II, Attachment II.7, Table F, (2010 Edition – Updated in 2012); anomalies are computed with respect to ERA-Interim-based climate (Courtesy of ECMWF web site <http://www.ecmwf.int>; Also see Bauer et al. 2015)

Ensemble forecasts: where are we today?

- Generating initial conditions: Each center has adopted their own approximate way of sampling from initial condition pdf.
 - Breeding (NCEP)
 - Singular vector (ECMWF)
 - Perturbed observation (Canada)
- Stochastic-dynamic ensemble work just beginning (e.g., Buizza et al. 1999)
- Many attempts to post-process ensemble forecasts to provide reliable probability forecasts.

....END

Thanks !

Introduction to Numerical Weather Prediction



Chapter 1: Introduction



Objective

- *To Describe The basic concept of NWP*
- *To Identify The NWP Processes And Components*
- *To List Weather Prediction Equation*

The basic concept of NWP

The basic concept of NWP

- ❖ The basic concept of NWP is to solve a set of partial differential equations (PDEs) that govern atmospheric motion and evolution (Kalnay 2003).
- ❖ The set of PDEs describes basic conservation laws, including the conservation of momentum, mass, energy, and water vapor.
- ❖ In order to predict the atmospheric state in the future, we must integrate this set of equations forward.
- ❖ Therefore, NWP is an initial value problem: given the current atmospheric conditions (initial conditions), we integrate the set of PDEs to obtain future atmospheric states.

The basic concept of NWP

- ❑ This initial value problem was defined early in the 1900s . Bjerknes (1904) stated that the ultimate problem in meteorology is weather forecasting (predicting future atmospheric conditions) and outlined an approach for tackling it.
- ❑ According to his approach, two conditions must be satisfied to successfully predict future atmospheric states
 - I. The present atmospheric conditions must be characterized as accurately as possible.
 - II. The basic laws, according to which the subsequent states develop out of the preceding ones, must be known.

NWP Processes And Components

Cont..

NWP Process involves three major phases

- 1. Analysis Phase (estimate present conditions)**
- 2. Prediction Phase (computer modeling)**
- 3. Post-Processing Phase (use of products)**

NWP Process

Gather
Observations



Data
Assimilation



Numerical
Weather
Predictions



Forecast
Postprocessing

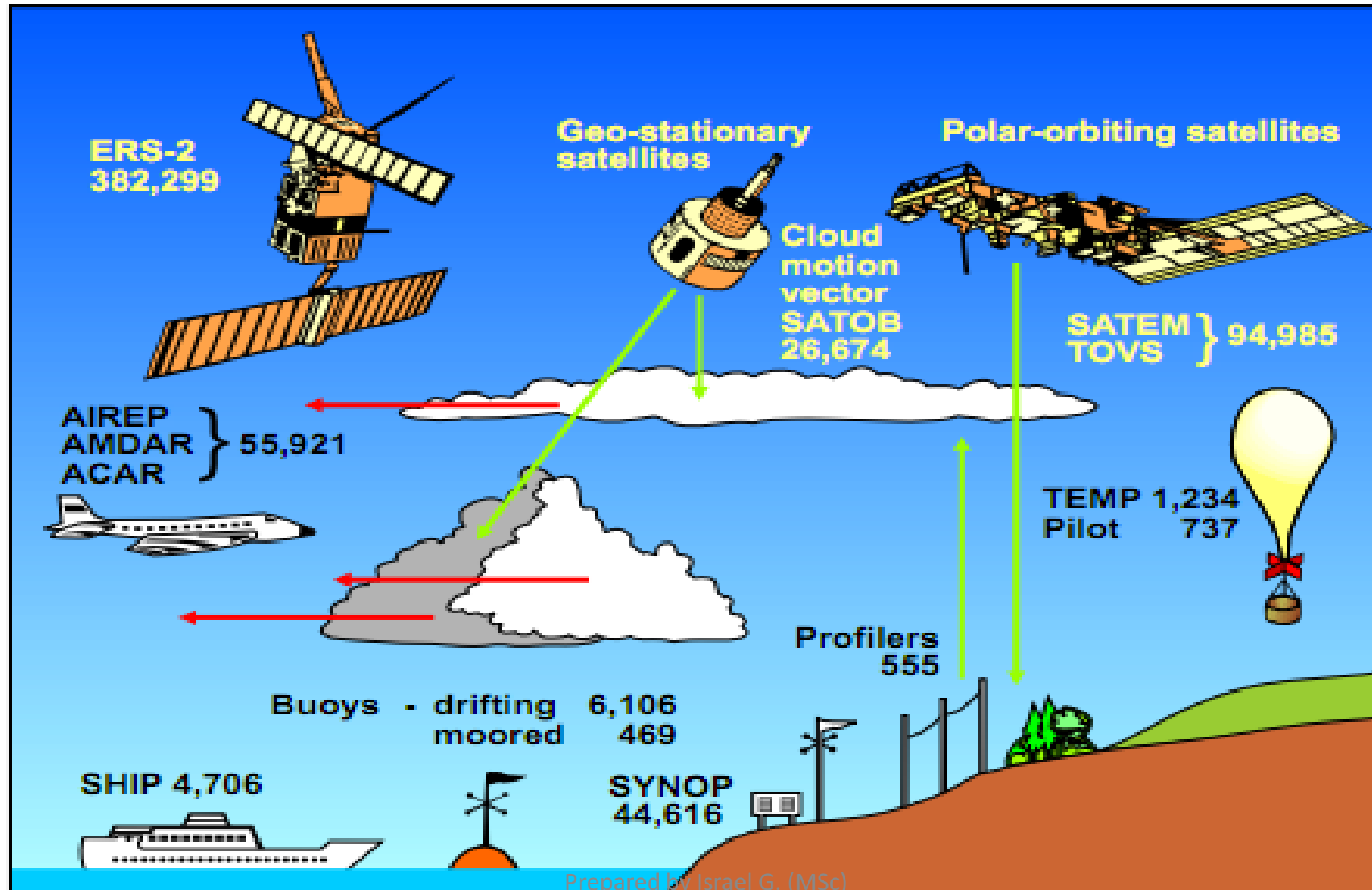


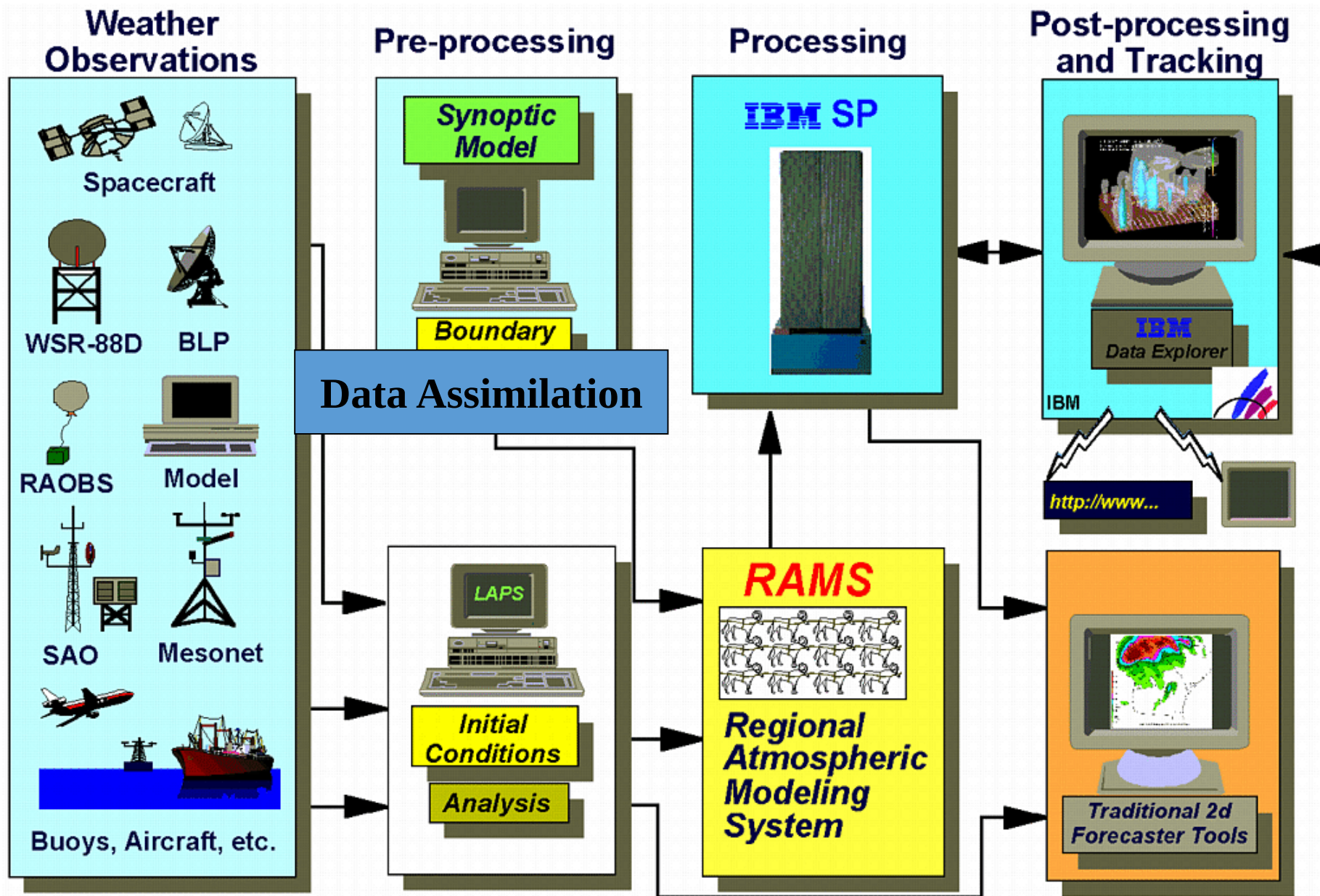
Issue forecasts,
Evaluate

Initial and Boundary Conditions

- Initial conditions define the atmosphere's current state...the starting point (depends on time)
 - **Time will be varied**
- Boundary conditions define the atmosphere's state and the domains' edges (depends on space)
 - **Space coordinate will be varied**

Type and number of observations used to estimate the atmosphere initial conditions

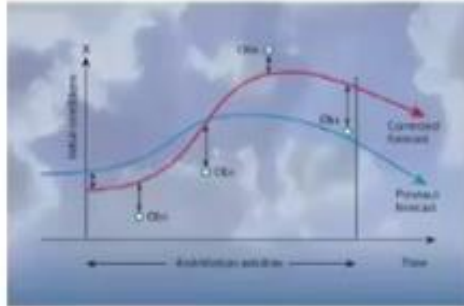




Observations



Data-assimilation



**Numerical weather
forecasts**



**Post-processing
and dissemination**



- The weather forecasting process involves:
 1. Collection of observations
 2. Using those observations to specify the initial conditions for the forecast
 3. The use of a model to extrapolate the state of the atmosphere forwards in time
 4. Assessment of the output of that model by experienced forecasters
 5. Production of forecasts for customers.

NWP model Components

1. Initial Conditions and Data Assimilation
2. Governing equations
3. Numerical Procedures
4. Boundary Conditions
5. Physical Processes
6. Model Output

Weather Prediction Equation

Weather Prediction Equation

- There is a complete set of seven equations with seven unknowns that governs the evolution of the atmosphere: *Newton's second law or conservation of momentum* (three equations for the three velocity components), *the continuity equation or conservation of mass*, *the equation of state for ideal gases*, *the first law of thermodynamics or conservation of energy*, and *a conservation equation for water mass*.
- In the Cartesian coordinate system, this set of equations can be written as follows:

$$\frac{d\mathbf{V}_3}{dt} = -2\boldsymbol{\Omega} \times \mathbf{V}_3 - \frac{1}{\rho} \nabla_3 p - \nabla_3 \Phi + \mathbf{F}, \text{ momentum equation,}$$

$$\frac{dT}{dt} = \frac{R}{C_p} \frac{T}{p} \frac{dp}{dt} + \frac{Q}{C_p}, \text{ thermodynamic equation,}$$

$$\frac{d\rho}{dt} = -\rho \nabla \cdot \mathbf{V}_3, \text{ continuity equation,}$$

$$\frac{dq}{dt} = M, \text{ water vapour equation,}$$

$$p = \rho R T, \text{ equation of state.}$$

The geopotential Φ is defined as the product of height z by acceleration due to gravity g , which combines only the effects of Newtonian gravity g^* and the centrifugal force (assuming that the Earth is isolated in space and so neglecting the effect of the other bodies of the solar system); it is expressed in J.kg^{-1} in SI units:

$$\Phi = g^* z - \frac{\Omega^2 r^2 \cos^2 \varphi}{2} = g z,$$

where Ω is the angular velocity of rotation of the Earth, r the radial distance ($r = \|\mathbf{r}\|$, where $\mathbf{r}$ is the radius vector as measured from the Earth's centre), and $r \cos \varphi$, the distance to the Earth's rotation axis at the latitude φ .

In equations (2.1) to (2.5), $\mathbf{V}_3$ represents the three-dimensional wind velocity, Ω the angular velocity vector of the Earth, ρ is air density, p pressure, T temperature, q specific humidity, R and C_p the perfect gas constant and specific heat at constant pressure for air. The symbol ∇_3 denotes the gradient operator, the symbol $\nabla_3 \cdot$ the divergence operator, and d/dt , the total derivative operator whose expression depends on the

- The above Equations are the equations of motion. They are also known as the primitive equations, because they forecast fundamental (primitive) variables rather than derived variables such as vorticity.
- To these equations we must add appropriate boundary conditions at the bottom and top of the atmosphere, then solve them using an integration process as suggested by Richardson (1922).

- The first six equations are **prognostic** (i.e., forecast the change with time), and the seventh (the ideal gas law) is **diagnostic** (not a function of time).
- These equations of motion are **nonlinear**, because many of the terms in these equations consist of products of two or more dependent variables. Also, they are **coupled** equations, because each equation contains variables that are forecast or diagnosed from one or more of the other equations. Hence, all 7 equations must be solved together.

Conservation of Momentum

Horizontal Momentum Equations

$$\underbrace{\frac{\partial u}{\partial t}}_{\text{time derivs.}} = - \underbrace{u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z}}_{\text{advection terms}} + \underbrace{\frac{uv \tan \phi}{a} - \frac{uw}{a}}_{\text{curvature terms}} - \underbrace{\frac{1}{\rho} \frac{\partial p}{\partial x}}_{\text{pres. grad.}} - \underbrace{2\Omega(w \cos \phi - v \sin \phi)}_{\text{Coriolis terms}} + \underbrace{F_x}_{\text{friction}}$$

$$\underbrace{\frac{\partial v}{\partial t}}_{\text{time derivs.}} = - \underbrace{u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z}}_{\text{advection terms}} - \underbrace{\frac{u^2 \tan \phi}{a} - \frac{uw}{a}}_{\text{curvature terms}} - \underbrace{\frac{1}{\rho} \frac{\partial p}{\partial y}}_{\text{pres. grad.}} - \underbrace{2\Omega u \sin \phi}_{\text{Coriolis terms}} + F_y$$

Advection: The transfer of heat or matter by the flow of a fluid, especially horizontally in the atmosphere or the sea.

ϕ = latitude, a = radius of the Earth, Ω = rotational frequency of Earth, F = friction

Conservation of Momentum

Vertical Momentum Equation

$$\underbrace{\frac{\partial w}{\partial t}}_{\text{time deriv.}} = -u \underbrace{\frac{\partial w}{\partial x} - v \frac{\partial w}{\partial y} - w \frac{\partial w}{\partial z}}_{\text{advection terms}} - \underbrace{\frac{u^2 + v^2}{a}}_{\text{curvature term}} - \underbrace{\frac{1}{\rho} \frac{\partial p}{\partial z}}_{\text{pres. grad.}} - \underbrace{2\Omega u \cos \phi}_{\text{Coriolis term}} - \underbrace{g}_{\text{gravity}} + \underbrace{F_z}_{\text{friction}}$$

ϕ = latitude, a = radius of the Earth, Ω = rotational frequency of Earth, F = friction

Thermodynamic Equation

$$\frac{\partial T}{\partial t} = -u \frac{\partial T}{\partial x} - v \frac{\partial T}{\partial y} + w(\gamma - \gamma_d) + \frac{1}{c_p} \frac{dH}{dt}$$

or, alternately...

$$\underbrace{\frac{\partial T}{\partial t}}_{\text{time deriv.}} = -u \frac{\partial T}{\partial x} - v \frac{\partial T}{\partial y} - w \frac{\partial T}{\partial z} - \underbrace{w\gamma_d}_{\text{dry adiabatic term}} + \underbrace{\frac{Q}{c_p}}_{\text{diabatic heating}}$$

advection terms
dry adiabatic term
diabatic heating

γ = lapse rate of temperature, γ_d = dry adiabatic lapse rate, Q = diabatic heating rate

Conservation of mass/Continuity Equations

(i.e., mass – top – and water vapor – bottom – are neither created nor destroyed)

$$\underbrace{\frac{\partial \rho}{\partial t}}_{\text{time derivs.}} = - \underbrace{u \frac{\partial \rho}{\partial x} + v \frac{\partial \rho}{\partial y} + w \frac{\partial \rho}{\partial z}}_{\text{advection terms}} - \underbrace{\rho \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right)}_{\text{divergence term}}$$
$$\underbrace{\frac{\partial q_v}{\partial t}} = - \underbrace{u \frac{\partial q_v}{\partial x} + v \frac{\partial q_v}{\partial y} + w \frac{\partial q_v}{\partial z}} + Q_v$$

q_v = water vapor mixing ratio, Q_v = source/sink of q_v due to phase changes

Cont..

Ideal Gas Law

$$p = \rho RT$$

- Seven equations; seven variables (u, v, w, ρ , p, T, q)

If you can solve the set of equations (often referred to as the *primitive equations/governing equations*) given in the past few slides, you can do NWP!

...but...

...how do we actually *solve* these equations???

Newton's Second Law (the momentum equations)

- The second law states that when a force does act upon the body, the change in momentum will be directly proportional to the force.
- In atmospheric terms, if we change the word 'body' to 'parcel of air' and if, additionally, we make the assumption that all parcels of air have a mass of 1 kg, then we have a law that allows us to calculate the acceleration of air parcels if we can quantify the forces acting up on them.
- This is of profound importance in weather forecasting. If we can calculate all the relevant forces acting on parcels of air then we can calculate any changes to the speed of movement of air parcels – essentially allowing us to forecast the wind speed.

- We can apply Newton's Second Law independently to motions in all three directions (west–east, south–north and vertically) in order to calculate accelerations in all three directions.

$$\sum F = a$$

- **Pressure gradient force.** This is the fundamental force in the atmosphere that sets air parcels in motion, or changes their momentum if they are already moving.
- Its magnitude is proportional to the local gradient of pressure and the force acts to accelerate parcels from higher pressure towards lower pressure. This includes vertical pressure gradients, which will always act to push air parcels vertically upwards.

- **Gravity.** Acting vertically downwards towards the Centre of the Earth.
- **Friction.** This force dissipates momentum at the Earth's surface due to interaction of air parcels with surface roughness elements, such as plants, trees, buildings and so on.
- The effects of surface friction are translated upwards through the lowest part of the atmosphere by shear stresses –that is, air that has been slowed down by friction at the surface acts to slow the layer of air flowing immediately above it.

Conservation of Mass

- **The Conservation of Mass** (often referred to as the principle of *continuity*). This law states that mass can be neither created nor destroyed.
- Following a parcel of air along its trajectory, the mass of that parcel, M , cannot be changed, although its shape and volume may vary.

Mathematically:

$$\frac{dM}{dt} = 0$$

First Law of Thermodynamics

- This law is a statement of the conservation of energy. Any heat added to a parcel of air is used either to increase the temperature, T , of the air parcel or to do work against the surrounding atmosphere as the parcel expands, or a combination of the two.
- This is often expressed meteorologically as:

$$Q = c_p \frac{dT}{dt} + \alpha \frac{dp}{dt}$$

where Q is heat, c_p is the specific heat capacity of the air and α is the specific volume of the air ($= 1/\rho$).

The equation of state for a perfect gas

- This is a *diagnostic* relationship between the fundamental thermodynamic variables – that is, there is no time derivative involved.
- Knowing any two thermodynamic quantities of a parcel of air (pressure, temperature, density, potential temperature etc.), the equation of state allows the calculation of any other thermodynamic variable. In meteorology the usual form of the equation of state is:

$$p = \rho RT$$

where R is the gas constant for dry air ($= 287 \text{ J kg}^{-1} \text{ K}^{-1}$).

Conservation of water

- Water vapour in the atmosphere is usually expressed in terms of a mixing ratio q (i.e. mass of water vapour per unit mass of dry air). Following an air parcel the conservation of water can be written as:

$$\frac{dq}{dt} = E - C$$

- where E refers to sources of water vapour (evaporation) and C refers to sinks (condensation). As with the First Law of Thermodynamics, the Lagrangian rate of change can be expanded into its various Eulerian components to give an expression in terms of changes in water vapour at a fixed point due to advection of water vapour by the wind field together with sources and sinks:

$$\frac{\partial q}{\partial t} = -u \frac{\partial q}{\partial x} - v \frac{\partial q}{\partial y} - w \frac{\partial q}{\partial z} + E - C$$

❑ **Conservation of water vapour.** Water can be present in the atmosphere in all three phases – liquid, gas or solid – and can change easily between them.

❑ Of course, water is being constantly lost from the atmosphere in the form of precipitation and gained through evaporation from the Earth's surface, so, in terms of weather forecast production, this is perhaps the main principle in the list where the source and sink terms are of most crucial importance.

- This complete set of equations clearly implies a set of fundamental *prognostic* variables which the NWP model will forecast.
- These variables are the three wind components u , v and w , a pressure-related variable (e.g. pressure on a height surface or *geopotential height* on a pressure surface), a mass-related variable (e.g. density), a temperature variable (e.g. T or θ) and a moisture related variable (e.g. water vapour mass mixing ratio q).
- NWP models can produce forecasts of many other variables (e.g. cloud amount, rainfall rates) but these seven are the bare minimum needed to advance the equations forward in time

....END

Thanks !